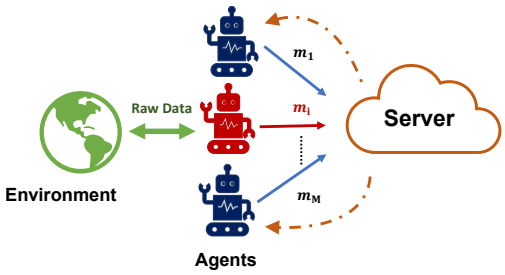
Robust Federated Q-Learning with Almost No Communication

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Problem and method. M agents interact with a common unknown MDP and collaborate through a server to estimate $Q^\star$. An omniscient ϵ fraction may send arbitrary, unbounded messages. The goal is to preserve collaborative sample-efficiency gains with little communication. **Robust Fed-Q** freezes Q_k throughout each H-sample epoch. Honest agents estimate transition kernels and form averaged Bellman-target tables for all state-action pairs. The server aggregates these tables coordinate-wise using median-of-means, updates the Q-table once, and broadcasts Q_{k+1}. As batches grow, honest messages concentrate, reducing statistical uncertainty and adversarial fluctuations.



Federated architecture.

Assumptions and notation. The finite MDP has S states, A actions, discount $\gamma \in (0,1)$, and deterministic rewards $|R(s,a)| \le \bar{R}$. Each honest agent receives $T = KH$ independent generative samples *per state-action pair*; honest streams are independent. Let $e_k := \|Q_k - Q^\star\|_\infty$ and $Q_0 = 0$.

Epoch update. Each honest agent estimates its transition kernel from H samples per pair and sends $d_{i,k} = \widehat{\mathcal{T}}_{i,k} Q_k$. The server computes

$$\widetilde{d}_k(s,a) = \mathrm{MoM}\{d_{i,k}(s,a)\}_{i=1}^{M}, \qquad Q_{k+1} = (1-\alpha)Q_k + \alpha \widetilde{d}_k.$$

1. Robust aggregation with concentrated honest messages. For failure probability $\delta \in (0,1)$, choose the number of MoM buckets as

$$P = \left\lceil 8\epsilon M + \tfrac{256}{7}\log(2SAT/\delta) \right\rceil, \qquad \bar{\delta} = \frac{\delta}{SAT}, \qquad \frac{M}{P} \ge 2.$$

The last condition requires a sufficiently small corruption fraction and enough agents. Batching reduces the spread of the honest targets, so their aggregation error shrinks with H even when adversarial messages are unbounded.

2. High-probability finite-time guarantee. For suitable c_1, set

$$K = \left\lceil \frac{c_1 \log(MT)}{1-\gamma} \right\rceil, \qquad \alpha = \frac{\log(MT)}{(1-\gamma)K}.$$

With probability at least $1-\delta$,

$$e_K \le \frac{e_0}{MT} + \mathcal{O}\left(\frac{\bar{R}\sqrt{\log(MT)}}{(1-\gamma)^{5/2}} \left[\sqrt{\frac{\log(2SAT/\delta)}{MT}} + \sqrt{\frac{\epsilon}{T}} \right] \right). \tag{1}$$

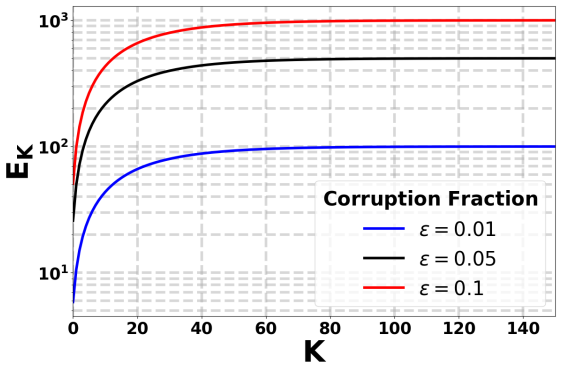
Without corruption, the dominant term is $\widetilde{\mathcal{O}}(1/\sqrt{MT})$, preserving the statistical benefit of collaboration.

3. Vanishing corruption effect and little communication. The adversarial contribution is $\widetilde{\mathcal{O}}(\sqrt{\epsilon/T})$ for fixed remaining parameters, so it vanishes as each honest agent acquires more data. The number of server communication rounds is

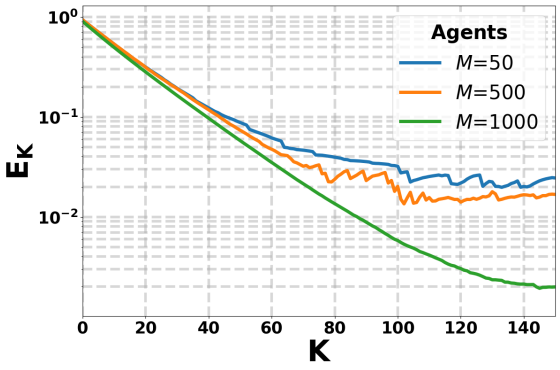
$$\text{rounds} = K = \mathcal{O}\left(\frac{\log(MT)}{1-\gamma} \right). \tag{2}$$

Each round uploads one $S \times A$ target table per agent and broadcasts the updated Q-table.

Numerical results. $S = 10$, $A = 5$, $\gamma = 0.5$, $R \in [0,1]$, $\alpha = 0.1$, $T = 25000$, and $\delta = 0.05$. Adversarial agents inject a 10^4 bias. Plots show $E_k = \|Q_k - Q^\star\|_\infty$; the robust stress test uses $\epsilon = 0.1$.



(a) Standard averaging is sensitive to corruption.



(b) Robust aggregation benefits from more agents.